

Solution Architecture

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Team ID	SWTID-2026-8550
Project Title	AI BlogNest API

3. Solution Architecture

The application exposes structured endpoints through a layered server architecture. Incoming requests are filtered before interacting with data services.

Architecture Diagram

API Client / Postman ↓ Express Server Gateway ↓ Authentication & Role Matrix Middleware
↓ Granular Services ✓ ↓ ↘ Blog Comments AI Flow ↓ Mongoose Database Interface
↓ MongoDB

Layer	Responsibilities
Interface	Postman/API clients query route paths.
Express Gateway	GET/POST/PUT/DELETE, JSON and CORS.
Auth & Role Matrix	Token validation and authorization.
Services	Blog lifecycle, comments and reporting rules.
Database Layer	Mongoose to MongoDB operations.
AI Flow	AI blog generation and summarization.
BlogGenerator	Automates blog drafting with AI.

ER/Data Relationships

User 1 — * Post User 1 — * Comment Post 1 — * Comment User: _id, name, email, hashed password, role. Post: _id, userID, name/title, photo, message, likes, timestamps.

MVC Pattern

Model: Mongoose schemas and MongoDB structures.

Controller: validates/delegates requests and returns JSON.
View/API Routing Layer: routes requests to controller handlers.